

An effective expression for the decomposition of Fano schemes of intersections of two quadrics

Pieter Belmans

September 30, 2026

This is the standalone version of Appendix A in Lena Ji, Fumiaki Suzuki: *Motivic classes of Fano schemes of lines on intersections of two quadrics* [3].

Abstract

The polynomials $M_{g,k,i}(t) \in \mathbb{Z}[t]$ introduced in [2] encode the multiplicities of Lefschetz twists of symmetric powers of a hyperelliptic curve in a conjectural motivic decomposition of Fano schemes of intersections of two quadrics. We show that these polynomials are effective, i.e., $M_{g,k,i}(t) \in \mathbb{Z}_{\geq 0}[t]$, settling the effectivity statement of the refined decomposition conjecture in loc. cit. We moreover give a manifestly effective closed expression for $M_{g,k,i}(t)$.

1 Introduction

Consider the polynomial in $\mathbb{Z}[t]$ introduced in [2, Equation (4)], given by

$$(1) \quad \begin{aligned} M_{g,k,i}(t) := & t^{i(g-k-1)} \left(\left[\begin{matrix} 2g-k-i \\ k+1-i \end{matrix} \right]_t - \left(t^{g-k-1} + t^{g+2k-3i} \right) \left[\begin{matrix} 2g-k-i-4 \\ k-i \end{matrix} \right]_t \right. \\ & - \left(t^{g-k} + t^{g-i} + t^{g+k-2i} + t^{3g-3k-4} + t^{3g-2k-4-i} + t^{3g-k-2i-4} \right) \left[\begin{matrix} 2g-k-i-4 \\ k-i-1 \end{matrix} \right]_t \\ & - \left(t^{3(g-k-1)} + t^{3(g-k-1)+1} + t^{3g-2k-i-3} + t^{3g-2k-i-2} \right) \left[\begin{matrix} 2g-k-i-4 \\ k-i-2 \end{matrix} \right]_t \\ & \left. - t^{4(g-k)-2} \left[\begin{matrix} 2g-k-i-4 \\ k-i-3 \end{matrix} \right]_t \right), \end{aligned}$$

where $g \geq 2$, $k = 0, \dots, g-2$ and $i = 0, \dots, k+1$. Here, we write

$$(2) \quad \left[\begin{matrix} n \\ m \end{matrix} \right]_t := \frac{(1-t^n)(1-t^{n-1}) \cdots (1-t^{n-m+1})}{(1-t)(1-t^2) \cdots (1-t^m)}$$

for the Gaussian binomial coefficient in the variable t , with integers $0 \leq m \leq n$, and the convention that $\left[\begin{matrix} n \\ m \end{matrix} \right]_t = 0$ for $m < 0$ or $m > n$.

Although the four subtractions in (1) might suggest otherwise, the polynomial $M_{g,k,i}(t)$ was expected to have non-negative coefficients, as expressed by the first half of [2, Conjecture B]. The following result confirms this expectation.

Theorem A. *For all $g \geq 2$, $k = 0, \dots, g-2$, and $i = 0, \dots, k+1$ the polynomial $M_{g,k,i}(t)$ defined in (1) is effective, i.e., $M_{g,k,i}(t) \in \mathbb{Z}_{\geq 0}[t]$.*

Moreover, with the convention that an empty product equals 1, it is given by the manifestly effective expression

$$(3) \quad M_{g,k,i}(t) = t^{i(g-k-1)} [x^{k+1-i}] \left(\prod_{j=0}^{g-k-3} \frac{1}{1-t^j x} \prod_{j=g-k+1}^{2g-2k-1} \frac{1}{1-t^j x} + t^{g-k-2} x \prod_{j=0}^{g-k-2} \frac{1}{1-t^j x} \prod_{j=g-k+2}^{2g-2k-1} \frac{1}{1-t^j x} \right),$$

where $[x^r]$ extracts the coefficient of x^r from the bracketed power series.

Here each factor $\frac{1}{1-t^j x} = \sum_{r \geq 0} t^{jr} x^r$ has non-negative coefficients, hence, the right-hand side of (3) is a non-negative combination of monomials in t .

Geometric motivation Let us briefly describe the geometric context for [2, Conjecture B]. For this geometric motivation, work over an algebraically closed field of characteristic zero. The variable t in (3) corresponds to the Lefschetz motive $\mathbb{L} = [\mathbb{A}^1]$, the class of the affine line in $K_0(\text{Var})$. For the smooth intersection of two quadrics $Q_1 \cap Q_2 \subset \mathbb{P}^{2g+1}$, with associated hyperelliptic curve C , [2, Conjecture B] predicts the identity

$$(4) \quad [\text{Fano}_k(Q_1 \cap Q_2)] = \sum_{i=0}^{k+1} M_{g,k,i}(\mathbb{L}) [\text{Sym}^i C] \quad \text{in } K_0(\text{Var}),$$

where $\text{Fano}_k(Q_1 \cap Q_2)$ denotes the Fano scheme of k -planes, and $\text{Sym}^i C$ the i th symmetric power. In other words, $M_{g,k,i}(\mathbb{L})$ plays the role of a motivic multiplicity, recording the multiplicity of each Lefschetz twist of $[\text{Sym}^i C]$ in (4). In [2], the effectivity was established only after setting $\mathbb{L} = 1$, where $M_{g,k,i}(1)$ is an explicit positive integer. In Corollary 7 we give a proof of this fact using our approach to Theorem A.

AI disclosure The effectivity part of [2, Conjecture B] was submitted as a benchmark problem to IMProofBench [4]. The initial proof of Theorem A was obtained by GPT-5.4 in March 2026. In order to turn the one-shot proof from IMProofBench into something digestible, this write-up was produced by the author, with LLMs used for proofreading.

Acknowledgements P.B. was partially supported by NWO (doi:10.61686/RZKLF82806). We thank the creators of IMProofBench for having built this benchmark system that led to the proof of Theorem A.

2 Proof

We will prove Theorem A in four steps:

1. First we perform an easy change of variables in Lemma 1, and encode the coefficients we are interested in using a generating function defined in (7).
2. Next, we recall a standard generating function for certain Gaussian binomial coefficients using Pochhammer symbols in Proposition 2, which is the main external input for the proof. In Lemma 3 we use this to give a new expression for (7).
3. In Lemma 4 and Corollary 5 we will further reduce this expression by elementary means.
4. Finally, in Lemma 6 we express the generating function in Corollary 5 as a sum of monomial multiples of reciprocals of products of Pochhammer symbols, making its effectivity explicit.

Reindexing First, a change of variables removes the dependence on i and turns the five-term sum into a single two-parameter family.

Lemma 1. *Set $a := g - k - 1$ (and thus $a \geq 1$) and $m := k - i$ (and thus $m \in \{-1, 0, \dots, k\}$). Then $M_{g,k,i}(t) = t^{ia} B_{a,m}(t)$, where*

$$(5) \quad \begin{aligned} B_{a,m}(t) := & \left[\begin{matrix} 2a + m + 2 \\ m + 1 \end{matrix} \right]_t - (t^a + t^{a+1+3m}) \left[\begin{matrix} 2a + m - 2 \\ m \end{matrix} \right]_t \\ & - (t^{a+1} + t^{a+1+m} + t^{a+1+2m} + t^{3a-1} + t^{3a-1+m} + t^{3a-1+2m}) \left[\begin{matrix} 2a + m - 2 \\ m - 1 \end{matrix} \right]_t \\ & - (t^{3a} + t^{3a+1} + t^{3a+m} + t^{3a+m+1}) \left[\begin{matrix} 2a + m - 2 \\ m - 2 \end{matrix} \right]_t \\ & - t^{4a+2} \left[\begin{matrix} 2a + m - 2 \\ m - 3 \end{matrix} \right]_t. \end{aligned}$$

Proof. Substitute $g = a + k + 1$ and $i = k - m$ into (1). We get

$$(6) \quad 2g - k - i = 2a + m + 2, \quad k + 1 - i = m + 1, \quad 2g - k - i - 4 = 2a + m - 2, \quad k - i - j = m - j,$$

and every monomial exponent becomes an affine function of a and m . Reading off the five terms of (1) gives (5). $\square$

For a fixed $a \geq 1$, and allowing k to vary so that we consider all integers $m \geq -1$, we assemble the polynomials $B_{a,m}(t)$ of Lemma 1 into the following generating function in a new variable x ,

$$(7) \quad F_a(t, x) := \sum_{m \geq -1} B_{a,m}(t) x^{m+1} \in \mathbb{Z}[t][[x]].$$

All admissible polynomials $M_{g,k,i}(t)$ are effective if and only if all polynomials $B_{a,m}(t)$ with $a \geq 1$ and $m \geq -1$ are effective, equivalently if $F_a(t, x) \in \mathbb{Z}_{\geq 0}[t][[x]]$ for every $a \geq 1$. Our goal is to give an explicit expression for $F_a(t, x)$, which we will accomplish in Corollary 5.

Packaging Gaussian binomial coefficients in a generating function For an integer $n \geq 0$ write the t -Pochhammer symbol

$$(8) \quad (y; t)_n := \prod_{r=0}^{n-1} (1 - t^r y) = (1 - y)(1 - ty) \cdots (1 - t^{n-1}y), \quad (y; t)_0 = 1.$$

Its inverse gives the following generating function of the Gaussian binomial coefficients along a fixed difference of indices, see [1, Theorem 3.3, Equation (3.3.7)].

Proposition 2. For every $n \geq 0$, as an identity of formal power series in y , we have that

$$(9) \quad \sum_{r \geq 0} \begin{bmatrix} n+r \\ r \end{bmatrix}_t y^r = \frac{1}{(y; t)_{n+1}}.$$

This is the tool that allows us to convert (5) from a difference of five terms using Gaussian binomials into a rational generating function. Note that (9) follows by summing the t -Pascal recursion $\begin{bmatrix} n \\ m \end{bmatrix}_t = \begin{bmatrix} n-1 \\ m \end{bmatrix}_t + t^{n-m} \begin{bmatrix} n-1 \\ m-1 \end{bmatrix}_t$. We apply this generating-function identity in Lemma 3; the subsequent proof of Lemma 4 consists of clearing denominators and collecting coefficients.

We use this to rewrite (7) as follows.

Lemma 3. For every $a \geq 1$,

$$(10) \quad F_a(t, x) = \frac{1}{(x; t)_{2a+2}} - \frac{t^a x}{(x; t)_{2a-1}} - \frac{t^{a+1} x}{(t^3 x; t)_{2a-1}} \\ - x^2 \left(\frac{t^{a+1} + t^{3a-1}}{(x; t)_{2a}} + \frac{t^{a+2} + t^{3a}}{(tx; t)_{2a}} + \frac{t^{a+3} + t^{3a+1}}{(t^2 x; t)_{2a}} \right) \\ - x^3 \left(\frac{t^{3a} + t^{3a+1}}{(x; t)_{2a+1}} + \frac{t^{3a+2} + t^{3a+3}}{(tx; t)_{2a+1}} \right) \\ - \frac{t^{4a+2} x^4}{(x; t)_{2a+2}}.$$

Proof. Multiply (5) by x^{m+1} and sum over $m \geq -1$. By Proposition 2, the leading term gives $1/(x; t)_{2a+2}$. For the remaining terms, the zero convention means that a summand with lower index $m - j$ contributes only for $m \geq j$. For $j = 0, 1, 2, 3$ and non-negative integers c, d , setting $r = m - j$ and applying Proposition 2 gives

$$(11) \quad \sum_{m \geq j} t^{c+dm} \begin{bmatrix} 2a+m-2 \\ m-j \end{bmatrix}_t x^{m+1} = t^{c+dj} x^{j+1} \sum_{r \geq 0} \begin{bmatrix} 2a+j+r-2 \\ r \end{bmatrix}_t (t^d x)^r \\ = \frac{t^{c+dj} x^{j+1}}{(t^d x; t)_{2a+j-1}}.$$

For example, the two parts of the second term of (5) contribute $-t^a x/(x; t)_{2a-1}$ and $-t^{a+1} x/(t^3 x; t)_{2a-1}$. $\square$

This allows us to prove that the following product of a Pochhammer symbol with the generating function $F_a(t, x)$ from (7) is an explicit polynomial.

Lemma 4. *For every $a \geq 1$ we have that*

$$(12) \quad (x; t)_{2a+2} F_a(t, x) = (1 - t^a x)(1 - t^{a+1} x)(1 - t^{2a+1} x^2).$$

Proof. Multiply (10) by $(x; t)_{2a+2} = \prod_{r=0}^{2a+1} (1 - t^r x)$. Every denominator divides this product, so clearing all of them leaves a polynomial of degree ≤ 4 in x . We will write $(x; t)_{2a+2} F_a(t, x) = T_0(t, x) + \dots + T_5(t, x)$, where

$$(13) \quad \begin{aligned} T_0(t, x) &= 1, \\ T_1(t, x) &= -t^a x(1 - t^{2a-1} x)(1 - t^{2a} x)(1 - t^{2a+1} x), \\ T_2(t, x) &= -t^{a+1} x(1 - x)(1 - tx)(1 - t^2 x), \\ T_3(t, x) &= -x^2 \left[(t^{a+1} + t^{3a-1})(1 - t^{2a} x)(1 - t^{2a+1} x) \right. \\ &\quad \left. + (t^{a+2} + t^{3a})(1 - x)(1 - t^{2a+1} x) + (t^{a+3} + t^{3a+1})(1 - x)(1 - tx) \right], \\ T_4(t, x) &= -x^3 \left[(t^{3a} + t^{3a+1})(1 - t^{2a+1} x) + (t^{3a+2} + t^{3a+3})(1 - x) \right], \\ T_5(t, x) &= -t^{4a+2} x^4. \end{aligned}$$

After expanding the products, the coefficient of each power of x is a sum of signed monomials t^{sa+c} . We group these contributions by their *slope* s , the coefficient of a in the exponent.

The coefficients of x^0 and x^1 are immediate, and equal 1 and $-(t^a + t^{a+1})$, respectively. For x^2 , the contribution $(t^{a+1} + t^{3a-1})(1 + t + t^2)$ from $T_1 + T_2$ cancels that from T_3 .

The coefficients of x^3 and x^4 are more involved, as monomials of several slopes contribute and partly cancel. For x^3 , the slopes 1, 3, and 5 occur before cancellation. The contributions of slopes 1 and 5 cancel, leaving only that of slope 3:

$$(14) \quad \begin{aligned} s = 1 : \quad & -(t^{a+2} + t^{a+3} + t^{a+4}) + t^{a+2} + (t^{a+3} + t^{a+4}) = 0, \\ s = 5 : \quad & -(t^{5a-1} + t^{5a} + t^{5a+1}) + (t^{5a-1} + t^{5a}) + t^{5a+1} = 0, \\ s = 3 : \quad & (t^{3a+1} + t^{3a+2}) + (t^{3a} + t^{3a+3}) + (t^{3a+1} + t^{3a+2}) \\ & - (t^{3a} + t^{3a+1}) - (t^{3a+2} + t^{3a+3}) = t^{3a+1} + t^{3a+2}, \end{aligned}$$

while for x^4 four slopes cancel and only $s = 4$ survives:

$$(15) \quad \begin{aligned} s = 1 : \quad & t^{a+4} - t^{a+4} = 0, \\ s = 3 : \quad & -(t^{3a+2} + t^{3a+3}) + (t^{3a+2} + t^{3a+3}) = 0, \\ s = 5 : \quad & -(t^{5a+1} + t^{5a+2}) + (t^{5a+1} + t^{5a+2}) = 0, \\ s = 7 : \quad & t^{7a} - t^{7a} = 0, \\ s = 4 : \quad & -t^{4a+2}. \end{aligned}$$

These five coefficients are exactly those of $(1 - t^a x)(1 - t^{a+1} x)(1 - t^{2a+1} x^2)$, which proves (12). $\square$

Corollary 5. *For every $a \geq 1$,*

$$(16) \quad F_a(t, x) = \frac{1 - t^{2a+1} x^2}{(x; t)_a (t^{a+2} x; t)_a}.$$

Proof. In (12) the factors $1 - t^a x$ and $1 - t^{a+1} x$ are the $r = a$ and $r = a + 1$ factors of $(x; t)_{2a+2}$; cancelling them leaves $1 - t^{2a+1} x^2$ over $\prod_{r=0}^{a-1} (1 - t^r x) \cdot \prod_{r=a+2}^{2a+1} (1 - t^r x) = (x; t)_a (t^{a+2} x; t)_a$. $\square$

Finishing the proof One elementary splitting of the numerator clears the last minus sign and exhibits $F_a(t, x)$ as a sum of two effective series.

Lemma 6. For every $a \geq 1$,

$$(17) \quad F_a(t, x) = \frac{1}{(x; t)_{a-1} (t^{a+2}x; t)_a} + \frac{t^{a-1}x}{(x; t)_a (t^{a+3}x; t)_{a-1}} \in \mathbb{Z}_{\geq 0}[t][[x]].$$

In particular $B_{a,m}(t) \in \mathbb{Z}_{\geq 0}[t]$ for all $a \geq 1, m \geq -1$.

Proof. Set $u = t^{a-1}x$ and $v = t^{a+2}x$. The identity $1 - uv = (1 - u) + u(1 - v)$ splits the numerator of (16) into terms divisible by factors already present in the denominator: the factor $1 - u$ is the last factor of $(x; t)_a$ and $1 - v$ is the first factor of $(t^{a+2}x; t)_a$, so cancelling them yields (17). Each remaining factor $1/(1 - t^jx) = \sum_{r \geq 0} t^{jr}x^r$ has non-negative coefficients. Hence $B_{a,m}(t) = [x^{m+1}]F_a(t, x) \in \mathbb{Z}_{\geq 0}[t]$. $\square$

We can now prove the main result of this appendix.

Proof of Theorem A. By Lemma 6, $B_{a,m}(t) \in \mathbb{Z}_{\geq 0}[t]$ for all $a \geq 1$ and $m \geq -1$. Hence $M_{g,k,i}(t) = t^{ia}B_{a,m}(t)$ lies in $\mathbb{Z}_{\geq 0}[t]$ by Lemma 1, which is the asserted effectivity.

For the explicit formula, expand each factor in (17) as a geometric series $1/(1 - t^jx) = \sum_{r \geq 0} t^{jr}x^r$ and extract the coefficient of x^{k+1-i} ; restoring the prefactor $t^{i(g-k-1)}$ then gives (3). $\square$

Setting $t = 1$ gives the sum of the coefficients of each polynomial $M_{g,k,i}(t)$. The following corollary recovers the formula for these sums established in [2, Lemma 3.17].

Corollary 7. Setting $t = 1$ in (3), we have $M_{g,k,k+1}(1) = 1$ and, for $i = 0, \dots, k$,

$$(18) \quad M_{g,k,i}(1) = \binom{2g-4-k-i}{k+1-i} + 2 \binom{2g-4-k-i}{k-i}.$$

Proof. At $t = 1$ each Pochhammer factor degenerates as $(y; t)_n \mapsto (1 - y)^n$, so by Corollary 5

$$(19) \quad F_a(1, x) = \frac{1 - x^2}{(1 - x)^{2a}} = \frac{1 + x}{(1 - x)^{2a-1}}.$$

The constant coefficient is 1, giving $M_{g,k,k+1}(1) = 1$. For $i = 0, \dots, k$, extracting the coefficient of x^{k+1-i} (with $a = g - k - 1$) gives $\binom{2g-k-i-3}{k+1-i} + \binom{2g-k-i-4}{k-i}$, which equals (18) by Pascal's identity. $\square$

Remark 8. The value $M_{g,k,i}(1)$ of Corollary 7 is obtained in [2, Lemma 3.17] by applying Pascal's identity repeatedly to the leading binomial of (1), but only *after* having set $t = 1$. For the polynomial $M_{g,k,i}(t) \in \mathbb{Z}[t]$ itself, a direct repetition of this argument does not seem to make effectivity apparent: the t -Pascal identity $\begin{bmatrix} n \\ m \end{bmatrix}_t = \begin{bmatrix} n-1 \\ m \end{bmatrix}_t + t^{n-m} \begin{bmatrix} n-1 \\ m-1 \end{bmatrix}_t$ introduces additional monomial weights whose cancellations must be controlled.

The remedy is to package the whole family into the generating function (7), and pass to its closed form. The t -Pascal recursion enters through the generating-function identity of Proposition 2, which gives (10). Clearing denominators and collecting coefficients then gives (12).

References

- [1] George E. Andrews. *The theory of partitions*. Cambridge Mathematical Library. Reprint of the 1976 original. Cambridge University Press, Cambridge, 1998, pp. xvi+255. ISBN: 0-521-63766-X. MR: 1634067.
- [2] Pieter Belmans, Jishnu Bose, Sarah Frei, Benjamin Gould, James Hotchkiss, Alicia Lamarche, Jack Petok, Cristian Rodriguez Avila, and Saket Shah. "On decompositions for Fano schemes of intersections of two quadrics". In: *Adv. Math.* 480.part C (2025), Paper No. 110506, 32. DOI: 10.1016/j.aim.2025.110506. MR: 4953422.
- [3] Lena Ji and Fumiaki Suzuki. *Motivic classes of Fano schemes of lines on intersections of two quadrics*. 2026. arXiv: 2609.36233v1 [math.AG].
- [4] Johannes Schmitt, Gergely Bérczi, Jasper Dekoninck, Jeremy Feusi, Tim Gehringer, et al. *IMProofBench: Benchmarking AI on Research-Level Mathematical Proof Generation*. 2026. arXiv: 2509.26076v2 [cs.CL].

Pieter Belmans, p.belmans@uu.nl

Mathematical Institute, Utrecht University, Budapestlaan 6, 3584CD Utrecht, Netherlands